

FPGA Security Challenges & Trends

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Like the 'horse of a different colour' in *Wizard of Oz*, field-programmable gate array (FPGA) and customisable system-on-chip (cSoC) integrated circuits are the chameleon workhorses of the electronics world. Available in a variety of different sizes and arrangements, these can be programmed to perform almost any logic function from small to large. The tightly coupled general-purpose microprocessor and associated sub-system of the cSoC can be programmed with firmware to perform tasks best suited to sequential execution. The ever expanding number of surrounding dedicated logic blocks, memories and interfaces such as math-blocks, SRAM, non-volatile memory and high-speed serialiser-deserialiser (SERDES) interfaces complete this almost infinitely versatile device. But, with all this flexibility, can a high standard of security be maintained?

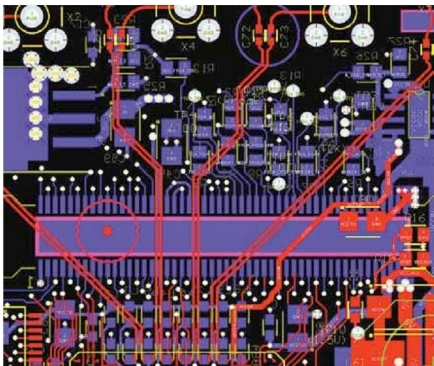
Before answering that question, let us first take up the question: What security objectives are we most likely interested about? Traditionally, FPGA and cSoC security is broken down into two distinct concepts: design security and data security.

Design security

In design security, the objective is to protect the interests of the IP owner—usually the original equipment manufacturer (OEM)—whose engineers designed the soft logic and firmware used to configure the device at the board- or system-level manufacturing step. Because of the investment in creating the FPGA or cSoC soft IP, the OEM wishes to keep its design confidential so that it can't be copied or cloned.

Possibly worse would be when the logic or firmware design is reverse-engineered, and the OEM's technology stolen and adapted for competitive

products. With all the flexibility offered by the use of FPGA, it is quite challenging to maintain a high standard of security in FPGA-based designs. Find out the design and data security concerns when working with FPGA-based designs, security features of different FPGA technologies and trends in advanced security architecture and countermeasures



FPGA-based design (Image courtesy: www.indiamart.com)

products. When this happens, the value of the company may be slashed to a fraction of its pre-theft value, as happened to one US company when its main customer stopped placing orders and started making the systems themselves, allegedly by stealing its design. In this example, the value of the US company's stock dropped 84 per cent.

Another design security concern is overbuilding. Overbuilding is where an unscrupulous contract manufacturer or his employees build more systems than the number authorised or received by the OEM, and then sell

the excess systems for their own profit. Aside from depressing the market and profits for the legitimate manufacturer, the OEM may also find customers of the often substandard illicit units coming to them for warranty repairs. The nightmare scenario is being sued for defects in a system that you didn't build but has your name written on it!

While it may be easy for these unscrupulous contract manufacturers to easily procure additional quantities of most components (including the FPGAs) in the system, because of the controls put in place by certain FPGA